

Xichen Zhang

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Education

Georgia Institute of Technology, Atlanta, GA

Master of Science in Electrical and Computer Engineering

Relevant Course: Biomedical Sensing and Imaging (ECE 6781), Advanced Digital Signal Processing (ECE 6250)

University at Buffalo, The State University of New York

Bachelor of Arts in Mathematics: Computing and Applied Concentration, May 2025

Relevant Courses: Introduction to Computer Science II, Survey of Differential Equations, Multivariable Calculus, Linear Algebra, Data Structures, Data-oriented Computing, Discrete Structures, Calculus III, Intro to Numerical Analysis I, Computer Complexity and Algorithms

Technical skills

Programming Languages: Python, SQL, Java, R, HTML, CSS

ML/AI Frameworks: TensorFlow, Keras, Scikit-learn, Numpy, Pandas

Work Experience

Teaching Assistant, CS department, University at Buffalo

Sept 2024 – May 2025

- Led weekly **2-hour review sessions** on algorithm analysis and complexity theory for a class of **200+ students**, enhancing conceptual understanding and bridging theoretical concepts with real-world problem solving
- Designed peer-based intervention to address learning disparities, later shared in internal pedagogy workshop

Security Analyst Intern, Merlin Dental Laboratory, Philadelphia

May 2024 – Aug 2024

- Engineered a **network security monitoring** dashboard using **Splunk**, which correlated data from **10+ security tools**; became the go-to resource for junior analysts during code deployment across the organization
- Conducted comprehensive security audits and penetration testing to identify vulnerabilities and **implement corrective actions**, improving the organization's overall security posture to be the most secure lab in the region

Academic Projects

Deep Learning for Detecting Genetic Interactions within Rare Disease: Harvard

Feb 2025 – Mar 2025

Rare Disease Hackathon– Genomic Diagnostics Track(Grand Champion)

- Expedited genomic data processing by creating an automated workflow that handled **100 datasets each week**, allowing researchers to focus on analysis rather than time-consuming data preparation tasks
- Developed a **deep learning model** to detect **genetic interactions (epistasis) in rare diseases**, leveraging advanced machine learning techniques to analyze complex genomic data

NoteWiz : UB AI Hackathon

Jan 2025 – Feb 2025

- Streamlined the organization of student notes using **AI and Streamlit**, enabling **200 students** to upload and classify **2,000+** notes within the first week of deployment, promoting efficient learning
- Revamped AI response mechanisms through prompt engineering and **retrieval-augmented generation (RAG)**, ensuring higher contextual relevance and earning recognition as team's most innovative intern

Dynamic System Research : Research Assistant – Buffalo, NY

Oct 2023 – May 2024

- Spearheaded the creation of **5 distinct computational models** to predict human responses to environmental changes, allowing for the identification of key factors influencing decision-making during crises
- Modeled long-term behavioral trends with computational methods, **processing 10,000+ data points** weekly and delivering findings that were presented to key stakeholders to inform future research directions

Professional Certification

- Google Cybersecurity
- Machine Learning Specialization(Stanford&Deeplearning.AI)

Extracurricular Activities

NetDef Cybersecurity Competition, *System Competitor, Buffalo, NY*

Apr 2023 – May 2023

- Earned **2nd** place in campus network defense systems against **14 teams**, eradicated invasive viruses in **Linux** and **Ubuntu** in a group of 4